

1 From TOP 2015 to TOP 2025: Updated Transparency Standards and a Tutorial for Creating
2 Synthetic Data

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Abstract

Journals increasingly require authors to share the data, code, and materials underlying published findings. The Transparency and Openness Promotion (TOP) Guidelines have provided a shared framework for formalizing these expectations since 2015, and a 2025 revision separates disclosure requirements and verification requirements and how each applies to methodological research. Both requirements pose a practical problem for authors who cannot share raw data due to consent restrictions, institutional review board limits, or third-party data use agreements. We describe the changes introduced in TOP 2025, outline how editorial policies can align with the revised guidelines, and provide a tutorial demonstrating synthetic data generation as a practical route to compliance when original data cannot be shared. We walk through an analysis using both the `synthpop` package in *R* and the `jaspSyntheticData` module in JASP, giving researchers a code-based and a point-and-click path to the same outcome. Analyses conducted on the synthetic datasets reproduced the pattern and magnitude of effects from the original data across both tools. Synthetic data generation offers authors a way to satisfy data-sharing expectations while protecting participant confidentiality, and pairing this tutorial with the TOP 2025 overview gives authors and reviewers a concrete reference for applying the updated standards.

Keywords: transparency and openness, synthetic data, reproducibility, TOP Guidelines, data sharing, open science

From TOP 2015 to TOP 2025: Updated Transparency Standards and a Tutorial for Creating Synthetic Data

Historically, *Behavior Research Methods* (BRM) has served as a venue for the dissemination of tools, analytic procedures, experimental paradigms, and computational workflows that shape how research is conducted and interpreted (Brysbaert et al., 2021). The journal’s emphasis on methodological innovation has made it an important outlet for research infrastructure, including software, measurement tools, and analytic approaches that can be implemented, evaluated, and extended across research domains. As empirical research has become increasingly computational and data-intensive, access to methodological components has become central to the evaluation of scientific claims. Understanding how findings were produced often requires access to not only written descriptions of methods but also to the materials, code, workflows, and analytic decisions underlying a study (Lee et al., 2018). When these components are unavailable or insufficiently documented, it becomes difficult to evaluate findings, reproduce analyses, or build upon prior work. In this context, transparency is not only a reporting journal preference, but a condition for interpreting and evaluating research. Given that methodological journals like BRM exist precisely to make these components available, transparency is not only a reporting preference for this outlet but a condition its editorial mission depends on and the same condition applies, in principle, to any journal whose claims rest on code, materials, or analytic decisions readers cannot otherwise inspect.

In response to these challenges, stakeholders (journals, funders, research institutes) find it necessary to clarify expectations for increased transparency in research reporting, tool development, and experimental procedures. The Transparency and Openness Promotion (TOP) Guidelines provided such a structure which allowed journals to formalize transparency standards (Nosek et al., 2015). Rather than defining transparency as a single requirement, the guidelines establish levels of implementation that distinguish among disclosure,

enforcement, and verification (Nosek et al., 2015). This structure allowed journals to clearly define expectations which are appropriate for their outlet and accommodate diverse methodological approaches in a consistent and interpretable manner. The recent revision of the TOP guidelines reflects a decade of experience implementing transparency standards across journals and disciplines and provides an opportunity to clarify how these standards apply within methodological research (Grant et al., 2026). BRM has long required exactly this kind of access to materials, code, and analytic workflows (Brysbaert et al., 2021), and we use it here as a working case for a re-alignment any journal adopting TOP 2025 will need to make. This tutorial provides an overview of the TOP Guidelines, explains the changes introduced in the 2025 revision, and describes how a journal’s editorial policies can be re-aligned with the updated guidelines. The goal is to clarify expectations for authors and reviewers and to support consistent application of transparency standards across journals adopting these guidelines.

The 2025 Update

Origins of TOP (2015). Following their introduction in 2015, the TOP Guidelines were adopted by journals, publishers, and research organizations across disciplines (Nosek et al., 2015). The guidelines established a shared framework for transparency standards at the journal level, organized around eight individual standards, such as data, analytic code, and materials transparency as shown in Table 1. Each of these standards were specified across four levels of increasing stringency, from no requirement (Level 0) through disclosure (Level 1), requirement (Level 2), and verification (Level 3) (Nosek et al., 2015). Journals selected a level for each standard independently, allowing an outlet to require, for example, that data be shared (Level 2) while treating preregistration as optional (Level 0).

Although the original development of the TOP Guidelines was multidisciplinary, the contributing authors were drawn primarily from the social and behavioral sciences (Nosek et al., 2015). The guidelines were, as a result, initially adopted most readily within research

environments sharing relatively similar methodological assumptions and infrastructure, before expanding across disciplines with substantially different research designs, data structures, and conventions for documentation and dissemination (Grant et al., 2026). This adoption demonstrated broad support for formalizing transparency standards at the journal level. Over time, adoption expanded to disciplines with substantially different research designs, data structures, and standards for documentation and dissemination (Grant et al., 2026). This broader application highlighted ambiguities in how transparency standards should be interpreted and operationalized across contexts.

Problems and Barriers of the 2015 TOP Guidelines. Editors and publishers reported difficulty translating endorsement of TOP 2015 into consistent editorial policy. Although many journals formally adopted the guidelines, fewer implemented requirements for enforcement or verification, and endorsement often functioned as a public commitment to transparency without corresponding changes to editorial procedures (Toomey et al., 2025). Two factors limited implementation. First, the guidelines did not always define levels precisely enough to support uniform interpretation across standards and disciplines; verification, in particular, lacked a uniform operational definition, limiting its feasibility as a broadly adopted editorial practice (Grant et al., 2026). Second, survey research with editors identified structural constraints including limited staffing, competing responsibilities, and insufficient infrastructure, that made policy revision alone insufficient, since implementing transparency requirements also demands procedures for evaluating compliance and managing exceptions (Naaman et al., 2023). Without dedicated infrastructure or precisely defined standards, disclosure requirements were more readily implemented than verification.

These experiences did not reflect opposition to transparency, but the difficulty of translating general principles into operational editorial standards. In response, the Center for Open Science coordinated a revision of the guidelines informed by feedback from editors, researchers, and other stakeholders (Grant et al., 2026), resulting in TOP 2025. The resulting update reflects accumulated experience with the guidelines and aims to clarify how

transparency standards can be defined and implemented within contemporary research and publication environments (Center for Open Science, 2025).

What Changed and Why. The 2025 revision introduces structural changes intended to clarify the relationship between transparency and verification (Grant et al., 2026). In the original structure, verification was incorporated as the highest level of implementation within individual transparency standards (Level 3, see Table 1). Experience with this structure demonstrated that disclosure and verification represent distinct editorial and methodological functions that benefit from explicit differentiation (Grant et al., 2026). The revised framework distinguishes between research practices and verification practices, and these guidelines are shown in Table 2. Research practices address the accessibility and documentation of methodological components, including data, code, materials, and analytic procedures. Verification practices address the extent to which research outputs can be independently evaluated using the available documentation and resources (Center for Open Science, 2025). This distinction clarifies that transparency facilitates verification but does not, by itself, ensure that verification has occurred.

This restructuring also reflects changes in research infrastructure and editorial workflows, and a direct comparison of the guidelines can be found in Table 3. The increasing availability of trusted repositories, shareable computational environments, and standardized documentation practices has expanded the feasibility of sharing and evaluating research components. The revised structure aligns transparency standards with these developments while maintaining flexibility across research domains. More broadly, the revision reflects an evolution in how transparency standards are conceptualized and implemented. Rather than representing a departure from the original guidelines, TOP 2025 refines and extends it in response to experience gained through widespread adoption and application (Grant et al., 2026).

Mapping Editorial Policy onto TOP 2025

BRM's emphasis on methodological transparency predates the TOP Guidelines. Long before formal endorsement, the journal already published and hosted supplementary code, materials, and data alongside articles (Buchanan, Valentine, & Maxwell, 2019; Vaughan, 2004), reflecting an expectation that published methods be specified in enough detail to permit independent implementation and evaluation. *BRM* converted this established practice into explicit editorial policy in 2021, when it formally adopted TOP Level 2 (Requirement) for the Data Transparency, Analytic Code Transparency, and Materials Transparency standards, requiring that articles include publicly available data, code, and materials, or an explanation of why these cannot be shared (Brysbaert et al., 2021; Nosek et al., 2015).

Under the restructured TOP 2025 guidelines, this same policy maps onto Level 2 (Shared and Cited) of the corresponding Research Practices of Data Transparency, Analytic Code Transparency, and Materials Transparency, which require that researchers deposit data, code, and materials in a trusted repository and cite that deposit within the article (Center for Open Science, 2025; Grant et al., 2026). A trusted repository, under this definition, is one that assigns a persistent identifier to each deposit, supports versioning so that later corrections or updates do not overwrite the original record, and commits to long-term preservation and public accessibility (Jantz & Giarlo, 2007). Registries such as re3data.org let authors confirm that a given platform meets these criteria before depositing materials. This includes general-purpose repositories such as FigShare and Zenodo, domain-specific archives appropriate to the data type, and a GitHub release paired with a Zenodo-issued DOI, provided the deposit satisfies the same persistence and versioning requirements.

What changes under TOP 2025 is not the requirement itself but its structure. The Level 2 policy still asks for the same thing it asked for in 2021, deposited data, code, and materials, and does not additionally require Level 3 (Certified), which would mean independent, third-party confirmation that the deposited materials meet best-practice

standards for their type. TOP 2025 simply makes that distinction explicit by separating Research Practices from Verification Practices (Table 3). The same logic applies to any exception for ethical or legal constraints, which TOP 2025 still requires authors to disclose even when full sharing is not possible (Level 1). When constraints such as participant confidentiality prevent depositing original data, authors are still expected to document what can be shared and why the remainder cannot be, and the tutorial that follows addresses exactly this case, showing how synthetic data generation lets authors satisfy the Level 2 data-sharing requirement without disclosing identifiable participant records.

Tutorial

Overview. One of the most significant barriers to data sharing under any transparency framework is the presence of sensitive, identifiable, or proprietary information in research datasets. Researchers may be unable to share raw data due to informed consent agreements, institutional review board restrictions, or data use agreements with third parties. However, the spirit of data transparency requirements is that others should be able to evaluate and reproduce the analyses underlying published claims. Synthetic data offer a solution to the problem, wherein researchers generate a new dataset that preserves the statistical structure of the original data without including the potentially identifiable original individual information (Jordon et al., 2022).

We demonstrate synthetic data generation using two tools that target different researcher skill sets and workflows. The **synthpop** package (Nowok, Raab, & Dibben, 2016) provides a code-based workflow for *R* (R Core Team, 2026) users who want programmatic control over the synthesis process and integration with existing analysis scripts. The **jaspSyntheticData** module (Buchanan, 2026) provides a point-and-click interface within JASP (JASP Team, 2026) for researchers who prefer a graphical environment. Both tools implement the same core methodology (i.e., **synthpop** is embedded into the JASP module), and produce comparable outputs, making them suitable alternatives depending on researcher

190 preference and infrastructure. Other *R* packages, such as *faux* (DeBruine, 2021), *simstudy*
191 (Goldfeld & Wujciak-Jens, 2020) may also be used, along with IVEware (Raghunathan,
192 Berglund, & Solenberger, 2018), which is stand alone simulation and imputation software.

193 Both tutorials use the same dataset and research question, allowing direct comparison
194 of the workflows. Each tutorial walks through the same sequence: preparing the data, fitting
195 the original analysis, generating the synthetic dataset, evaluating its utility, and confirming
196 that the synthetic data support the same scientific conclusions. Because synthesis introduces
197 variability at each modeling step, synthetic data are not expected to reproduce the original
198 results exactly. Instead, the goal is a synthetic dataset whose analyses approximate the
199 original's estimates and pattern of significance closely enough to support the same
200 substantive conclusions.

```

# The data package is available on GitHub; install it once with:
# install.packages("remotes")
# remotes::install_github("marton-balazs-kovacs/trolleyMultilabReplication")
library(trolleyMultilabReplication)
library(synthpop)

data(trolley)

# Compute Oxford Utilitarianism composite
util_cols <- paste0("oxford_utilitarian_", 1:9)
trolley$util_score <- rowMeans(trolley[, util_cols], na.rm = TRUE)

# Select analytic variables and restrict to Male/Female for Gender
df <- trolley[, c("trolley_1_rate", "util_score", "Region", "Age", "Gender")]
df$Gender <- factor(df$Gender, levels = 1:2, labels = c("Male", "Female"))
df$Region <- factor(df$Region)
df <- df[complete.cases(df), ]

```

Example Data. We used R (Version 4.5.3; R Core Team, 2026) and the R-packages *papaja* (Version 0.1.4; Aust & Barth, 2025), *synthpop* (Version 1.9.2; Nowok et al., 2016), *tinylabels* (Version 0.2.5; Barth, 2025), and *trolleyMultilabReplication* (Version 0.1.0; Kovacs, 2026) for all analyses in this tutorial.

To illustrate the workflow, we use data from the Psychological Science Accelerator's Trolley Problem Multilab Replication (Bago et al., 2022; Moshontz et al., 2018), available at <https://github.com/bencebago/PSA-006-Trolley-problem>. The dataset contains responses

from 27502 participants across the globe who completed a set of moral dilemma scenarios based on the original trolley problem (Thomson, 1976), along with measures of moral attitudes. For this tutorial, we analyze the following research question: Do utilitarian attitudes predict how morally acceptable people find the trolley dilemma, and does this relationship vary by world region?

We focus on five variables:

- **trolley_1_rate**: Perceived moral acceptability of the “footbridge pole” dilemma (1 = completely unacceptable, 9 = completely acceptable)
- **util_score**: A composite score derived from the nine-item Oxford Utilitarianism Scale (Kahane, Everett, Earp, Farias, & Savulescu, 2018), computed as the item mean (1 = strongly disagree, 7 = strongly agree; higher scores indicate stronger utilitarian attitudes)
- **Region**: Participant world region (Eastern, Southern, Western)
- **Age**: Participant age in years
- **Gender**: Participant gender (Male, Female; participants selecting other options are excluded to maintain a clean two-level factor for this demonstration)

After selecting complete cases on all five variables, the analytic sample contains 3495 participants.

Original Analysis. We fit a linear regression predicting moral acceptability ratings from utilitarian attitudes, world region, their interaction, age, and gender. This model tests whether the relationship between utilitarian reasoning and moral acceptability is consistent across regions or whether it differs in strength or direction depending on cultural context.

```
m_orig <- lm(trolley_1_rate ~ util_score * Region + Age + Gender, data = df)
summary(m_orig)
```

##

```

232 ## Call:
233 ## lm(formula = trolley_1_rate ~ util_score * Region + Age + Gender,
234 ##     data = df)
235 ##
236 ## Residuals:
237 ##      Min       1Q   Median       3Q      Max
238 ## -4.9117 -1.5329 -0.2648  1.3396  8.2745
239 ##
240 ## Coefficients:
241 ##              Estimate Std. Error t value Pr(>|t|)
242 ## (Intercept)      0.618083   0.282648   2.187 0.028826 *
243 ## util_score        0.931464   0.072645  12.822 < 2e-16 ***
244 ## RegionSouthern    0.577997   0.362135   1.596 0.110562
245 ## RegionWestern    0.910511   0.357180   2.549 0.010840 *
246 ## Age              -0.016481   0.003038  -5.424 6.22e-08 ***
247 ## GenderFemale     -0.247007   0.072271  -3.418 0.000639 ***
248 ## util_score:RegionSouthern -0.170993   0.096209  -1.777 0.075604 .
249 ## util_score:RegionWestern -0.189952   0.096188  -1.975 0.048369 *
250 ## ---
251 ## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
252 ##
253 ## Residual standard error: 1.995 on 3487 degrees of freedom
254 ## Multiple R-squared:  0.1276, Adjusted R-squared:  0.1258
255 ## F-statistic: 72.84 on 7 and 3487 DF,  p-value: < 2.2e-16

```

256 Utilitarian attitude scores were a positive predictor of moral acceptability ratings
 257 ($b = 0.93$, 95% CI [0.79, 1.07], $t(3487) = 12.82$, $p < .001$), indicating that participants with
 258 higher utilitarian orientations rated the trolley dilemma as more morally acceptable. Age

was negatively associated with acceptability ratings ($b = -0.02$, 95% CI $[-0.02, -0.01]$, $t(3487) = -5.42$, $p < .001$), and female participants gave lower acceptability ratings than male participants ($b = -0.25$, 95% CI $[-0.39, -0.11]$, $t(3487) = -3.42$, $p < .001$). The interaction between utilitarian attitudes and region was significant for Western participants relative to Eastern participants ($b = -0.19$, 95% CI $[-0.38, 0.00]$, $t(3487) = -1.97$, $p = .048$), suggesting the relationship between utilitarianism and acceptability was different in Western samples. The model explained approximately 13% of variance in moral acceptability ratings, $R^2 = .13$, 90% CI $[0.11, 0.14]$, $F(7, 3487) = 72.84$, $p < .001$.

Generating Synthetic Data. Next, we generated a synthetic version of the analytic dataset using `synthpop`. The `syn()` function accepts a dataframe and creates each variable in sequence using classification and regression trees (CART) for non-numeric variables and continuous variables, respectively. CART-based synthesis is a flexible, nonparametric approach that can capture nonlinear relationships and interactions without requiring distributional assumptions (Berk, 2008; Drechsler & Reiter, 2011).

```
syn_out <- syn(df, seed = 42)
```

The `syn()` call returns a `synds` object containing the synthetic dataset (`syn_out$syn`), the synthesis specifications, and methods used for each variable. A summary of the object lists the method applied to each variable.

```
summary(syn_out)
```

```
## Synthetic object with one synthesis using methods:
```

```
## trolley_1_rate    util_score      Region      Age      Gender
##      "sample"      "cart"      "cart"      "cart"      "cart"
```

```
##
```

```
## trolley_1_rate    util_score      Region      Age      Gender
```

```
## Min.      :1.000    Min.      :1.000    Eastern : 899    Min.      :18.00    Male      :1240
```

```

282 ## 1st Qu.:2.000 1st Qu.:3.111 Southern:1262 1st Qu.:20.00 Female:2255
283 ## Median :3.000 Median :3.667 Western :1334 Median :22.00
284 ## Mean :3.504 Mean :3.686 Mean :27.01
285 ## 3rd Qu.:5.000 3rd Qu.:4.222 3rd Qu.:28.00
286 ## Max. :9.000 Max. :6.889 Max. :90.00

```

287 ***Evaluating Utility.*** Before using the synthetic data for analysis, we should verify
 288 that the synthetic dataset faithfully reproduces the distributional properties of the original.
 289 `synthpop` provides the `compare()` function for this purpose, which generates side-by-side
 290 comparisons of variable distributions and computes the propensity score mean squared error
 291 (pMSE) as a global metric (Snoke, Raab, Nowok, Dibben, & Slavkovic, 2018; Woo, Reiter,
 292 Oganian, & Karr, 2009). The pMSE measures how well a classifier can distinguish real from
 293 synthetic observations wherein values close to zero indicate high utility (the synthetic data
 294 are difficult to distinguish from the original), while large values indicate a poor match.

```

# Save the comparison; a 2 x 3 grid keeps all five variables in one figure
syn_compare <- compare(syn_out, df, nrow = 2, ncol = 3, print.flag = FALSE)

# Utility statistics for each variable
syn_compare$tab.utility

```

```

295 ##                pMSE    S_pMSE df
296 ## trolley_1_rate 1.457150e-05 0.2716128 3
297 ## util_score     7.520982e-06 0.1051433 4
298 ## Region         1.681477e-05 0.4701409 2
299 ## Age            6.192843e-05 0.8657594 4
300 ## Gender         1.503973e-05 0.8410217 1

```

```
syn_compare$plots
```

The pMSE values for all five variables were near zero (range: < 0.001 to < 0.001), and the standardized pMSE ratios (S_{pMSE}) were all less than 1.0, indicating that the synthetic data achieve good utility across all variables. Visual inspection of the distribution plots confirms that the synthetic distributions closely match the originals for both continuous variables (utilitarian attitudes, age, acceptability ratings) and categorical variables (region, gender). See Figure 1 for example output.

We also assess analysis-specific utility by computing a measure of how well the synthetic data reproduce the regression coefficients from the original model. This value can be calculated using `utility.gen()` for general utility, or by directly comparing model estimates as we do in the next section.

Replicating the Analysis on Synthetic Data. The strongest test of synthetic data utility for scientific purposes is whether the analysis conducted on the synthetic data leads to the same conclusions as the analysis on the original data. We fit the same regression model to the synthetic dataset and compare coefficient estimates.

```
syn_df <- syn_out$syn
m_syn <- lm(trolley_1_rate ~ util_score * Region + Age + Gender, data = syn_df)
summary(m_syn)
```

```
##
```

```
## Call:
```

```
## lm(formula = trolley_1_rate ~ util_score * Region + Age + Gender,
```

```
##      data = syn_df)
```

```
##
```

```
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
```

```

322 ## -4.9029 -1.4896 -0.2657 1.3227 7.2987
323 ##
324 ## Coefficients:
325 ## Estimate Std. Error t value Pr(>|t|)
326 ## (Intercept) 0.227761 0.280182 0.813 0.416330
327 ## util_score 0.991064 0.071530 13.855 < 2e-16 ***
328 ## RegionSouthern 0.646502 0.353369 1.830 0.067404 .
329 ## RegionWestern 0.880698 0.351863 2.503 0.012362 *
330 ## Age -0.011350 0.002949 -3.849 0.000121 ***
331 ## GenderFemale -0.157298 0.071990 -2.185 0.028956 *
332 ## util_score:RegionSouthern -0.205964 0.094282 -2.185 0.028988 *
333 ## util_score:RegionWestern -0.182576 0.094483 -1.932 0.053396 .
334 ## ---
335 ## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
336 ##
337 ## Residual standard error: 1.976 on 3487 degrees of freedom
338 ## Multiple R-squared: 0.1418, Adjusted R-squared: 0.1401
339 ## F-statistic: 82.31 on 7 and 3487 DF, p-value: < 2.2e-16

```

The coefficient estimates from the synthetic data closely tracked those from the original data (see Table 4). The main effects of utilitarian attitudes, Western region, age, and gender were statistically significant in both analyses. The effect of utilitarian attitudes was reproduced with high fidelity (original: $b = 0.93$, 95% CI [0.79, 1.07]; synthetic: $b = 0.99$, 95% CI [0.85, 1.13]), as were the age effect (original: $b = -0.02$, 95% CI [-0.02, -0.01]; synthetic: $b = -0.01$, 95% CI [-0.02, -0.01]) and the gender effect (original: $b = -0.25$, 95% CI [-0.39, -0.11]; synthetic: $b = -0.16$, 95% CI [-0.30, -0.02]). The two interaction terms were where the analyses diverged. The interaction between utilitarian attitudes and Western region was nearly identical in size but was not significant in the synthetic data (original:

$b = -0.19$, 95% CI $[-0.38, 0.00]$, $t(3487) = -1.97$, $p = .048$; synthetic: $b = -0.18$, 95% CI $[-0.37, 0.00]$, $t(3487) = -1.93$, $p = .053$), whereas the interaction for Southern region was not significant in the original data but was significant in the synthetic data (original: $b = -0.17$, 95% CI $[-0.36, 0.02]$, $t(3487) = -1.78$, $p = .076$; synthetic: $b = -0.21$, 95% CI $[-0.39, -0.02]$, $t(3487) = -2.18$, $p = .029$). The small differences introduced by synthesis were enough to change whether each interaction was significant, even though the estimates themselves changed little. Synthetic data reproduced the size and direction of the effects well, so researchers should compare estimates and confidence intervals rather than expect identical significance decisions. The proportion of explained variance was similar across datasets (original: $R^2 = .13$, 90% CI $[0.11, 0.14]$; synthetic: $R^2 = .14$, 90% CI $[0.12, 0.16]$).

`synthpop` also provides a formal inference procedure for synthetic data analyses via `lm.synds()`, which adjusts standard errors to account for the additional uncertainty introduced by synthesis. For single-synthesis workflows (the default), this function implements the correction described by Raab, Nowok, and Dibben (2018). Researchers who require formal statistical inference from synthetic data should use this approach.

```
fit_synds <- lm.synds(trolley_1_rate ~ util_score * Region + Age + Gender,
                      data = syn_out)
summary(fit_synds)
```

```
## Fit to synthetic data set with a single synthesis. Inference to coefficients
## and standard errors that would be obtained from the original data.
##
## Call:
## lm.synds(formula = trolley_1_rate ~ util_score * Region + Age +
##          Gender, data = syn_out)
##
## Combined estimates:
```

```

372 ##                xpct(Beta) xpct(se.Beta) xpct(z) Pr(>|xpct(z)|)
373 ## (Intercept)      0.2277605    0.2801823  0.8129    0.4162747
374 ## util_score      0.9910642    0.0715300 13.8552    < 2.2e-16 ***
375 ## RegionSouthern  0.6465018    0.3533685  1.8295    0.0673188 .
376 ## RegionWestern  0.8806982    0.3518635  2.5030    0.0123161 *
377 ## Age             -0.0113503    0.0029492 -3.8486    0.0001188 ***
378 ## GenderFemale    -0.1572976    0.0719899 -2.1850    0.0288892 *
379 ## util_score:RegionSouthern -0.2059641    0.0942818 -2.1846    0.0289213 *
380 ## util_score:RegionWestern -0.1825759    0.0944833 -1.9324    0.0533149 .
381 ## ---
382 ## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

383 ***Practical Guidance for Sharing Synthetic Data.*** Synthetic data generated
384 with `synthpop` can be deposited in a public repository in place of, or alongside, the original
385 data. Synthetic data are most appropriate when the original data cannot be shared due to
386 participant confidentiality, data use restrictions, or consent limitations but when researchers
387 nonetheless wish to enable others to verify analytic code, explore the data structure, or
388 conduct methods development. Synthetic data are less appropriate as a substitute for
389 original data in meta-analyses or secondary analyses where individual-level heterogeneity is
390 the scientific target.

391 Although synthetic data contain no original records, synthesis methods that too closely
392 reproduce unusual combinations of values in small subgroups may pose residual disclosure
393 risk. `synthpop` includes the `disclosure()` function for evaluating identity and attribute
394 disclosure risk before releasing a synthetic dataset. Researchers should run this check as a
395 standard step in the workflow and consider increasing the smoothing (e.g., by adjusting the
396 `minbucket` argument in CART synthesis) if disclosure risk exceeds acceptable thresholds for
397 their context.

Shared synthetic datasets should be accompanied by the *R* script used to generate them (including the random seed), the **synthpop** version, and a brief statement clarifying that the data are synthetic and do not contain original participant records. This documentation allows others to trace the properties of the synthetic data back to the synthesis specifications. By default, **syn()** generates a single synthetic dataset. For formal statistical inference on synthetic data, generating multiple synthetic datasets and combining estimates using Rubin’s combining rules should reduce the additional uncertainty introduced by any single synthesis run.

Tutorial: JASP (jaspSyntheticData**).** The **jaspSyntheticData** module (Buchanan, 2026) provides a synthesis workflow through JASP’s graphical interface, making synthetic data generation accessible to researchers who work outside of *R* or who prefer a menu-driven environment. The module is available as a side module within JASP and uses **synthpop** for the sequential synthesis, with two additional steps. The module generates five synthetic datasets, selects one at random, and rescales numeric variables within each category so that their means and standard deviations match the original data. Because of these steps, the synthetic data produced in JASP will not be identical to those produced by the *R* workflow, even with the same random seed. The output includes a preview of the synthetic data, utility statistics, and distribution comparison plots similar to those created with the *R* workflow described above, and the synthetic dataset can be saved as a CSV file. This section walks through the same analytic example using JASP, beginning from data import and ending with an exported synthetic dataset ready for sharing. This tutorial was created with JASP version 0.95.5 (JASP Team, 2026) and version 0.2.0 of the **jaspSyntheticData** module (Buchanan, 2026).

Installing and Opening the Module. The **jaspSyntheticData** module is installed from within JASP via the module menu. Click the “+” icon in the top right corner, select the Community option under Channel, and scroll down to Synthetic Data. After installing, the module appears in the top menu bar alongside other JASP analyses. Users

who do not yet have JASP installed can download it from jasp-stats.org; the software is free and available for Windows, macOS, and Linux.

Loading the Data. The same analytic dataset used in the *R* tutorial and loaded into JASP using File > Open. JASP automatically detects variable types on import; continuous variables will be recognized as scale variables and categorical variables as nominal or ordinal. Variable types can be adjusted by clicking the variable type icon in the column header if JASP’s automatic detection does not match the intended measurement level.

Setting Up the Synthesis. Open the `jaspSyntheticData` module from the top menu. The main analysis panel contains a variable assignment box and a set of synthesis options. Move all five analytic variables into the Variables box. The synthesis order follows the order of the variables in the Variables box, which mirrors the `synthpop` default behavior. The first variable is sampled from its observed values, and each subsequent variable is synthesized conditional on all variables above it in the list. Variables can be reordered within the box if a different synthesis sequence is preferred.

The Options section contains the synthesis settings. Set a random seed so the synthesis is reproducible. The row count can stay at its default, which generates the same number of rows as the original data. The synthesis method defaults to CART, which matches the *R* tutorial and is appropriate for datasets containing a mix of continuous and categorical variables; conditional trees and parametric models are also available. The analysis runs automatically once variables are assigned and updates whenever an option changes.

Reviewing Synthesis Output. Results appear in the output panel on the right side of the screen. The output includes a table showing whether each variable was treated as continuous or categorical, a preview of the first 10 rows of the synthetic data, a table of utility statistics for each variable, and distribution comparison plots.

The distribution comparison plots are produced by the same `compare()` function used in the *R* tutorial. Each panel shows the percentage of observations in each category, or in

each range of values for continuous variables, with the original data in dark blue and the synthetic data in light blue. Close correspondence between the paired bars indicates good utility.

The Utility by Variable table reports the pMSE and standardized pMSE (S_{pMSE}) for each variable, computed on the final synthetic dataset after all adjustments. As with the *R* output, pMSE values near zero and S_{pMSE} values near or below 1 indicate that the synthetic data are difficult to distinguish from the original. The Utility section also offers an Overall utility option, which fits a single model using all variables at once to check whether the relationships between variables are preserved; this option is off by default because it can be slow for large datasets. Users should review both the visual comparisons and the utility statistics before proceeding to share or use the synthetic data.

Replicating the Analysis on Synthetic Data. To verify that the synthetic data support the same scientific conclusions as the original, the regression analysis can be run on the synthetic dataset in JASP. The synthetic data are not added to the open data file, so they first need to be saved and opened as a new dataset. In the Save synthetic dataset section, type or browse to a file path (e.g., `synthetic-data.csv`) and press Enter to write the CSV file. Open the saved file in JASP and navigate to Regression > Linear Regression. Assign `trolley_1_rate` as the dependent variable, `util_score` and `Age` as covariates, and `Region` and `Gender` as factors, then add the `util_score × Region` interaction in the Model section. The supplemental files include the exported JASP results (`.html`) so readers can inspect the analysis settings and output.

The regression coefficients from the synthetic data should be similar in direction and size to those from the original analysis (see Table 5). Because the module applies additional processing steps and draws a different synthetic dataset, the estimates will not exactly match the synthetic results from the *R* tutorial. As in the *R* tutorial, the interaction terms differed in the significance decision at $\alpha < .05$, as the interaction between utilitarian attitudes and

Western region was significant in the original data but not in the JASP synthetic data. Small differences in precise values are expected, so the comparison should focus on the size and direction of the estimates rather than on whether each one is significant.

Practical Guidance for Sharing Synthetic Data. The workflow for sharing JASP-generated synthetic data follows the same principles described for the *R* workflow. Export the synthetic dataset from JASP as a CSV file and deposit it in a public repository (e.g., Zenodo, figshare, or a domain-specific archive). Accompanying documentation should include the JASP version used, the `jaspSyntheticData` module version, the random seed and synthesis method, and a statement clarifying that the file contains synthetic rather than original data.

Researchers should not share the JASP analysis file (`.jasp`) for this purpose, because it stores a full copy of the dataset that was loaded, including the original data used for synthesis. Instead, the analysis record can be shared by exporting the results as an `.html` file (File > Export Results). The exported file contains the output tables and plots along with the analysis settings (e.g., the variables, seed, and synthesis options), but no raw data. Depositing this `.html` file alongside the synthetic CSV gives readers a record of how the synthetic data were produced and evaluated without exposing the original data.

Choosing a Trusted Repository. Under TOP 2025, sharing synthetic data counts toward Data Transparency only when the files are deposited in a trusted repository, which is one that assigns a persistent identifier, supports versioning, and commits to long-term preservation (see the TOP 2025 section above). General-purpose repositories such as Zenodo, figshare, Dryad, and Harvard Dataverse meet these criteria and accept any file type, which makes them a practical default for a synthetic CSV, the synthesis script, and the exported JASP results. Domain-specific archives are another option when one exists for the data type (e.g., OpenNeuro for neuroimaging data), and social science archives such as the Inter-university Consortium for Political and Social Research (ICPSR) are particularly useful for synthetic data workflows because they offer controlled-access deposits. The original data

can be archived under restricted access while the synthetic version is released publicly under the same study record. Whichever repository is chosen, the deposit should be cited in the article using its persistent identifier, and any later updates to the synthetic data should be released as new versions of the same record rather than as separate uploads.

Researchers who are unsure whether a platform qualifies, or who are looking for a repository suited to their field, can use the Registry of Research Data Repositories (re3data.org). From the Search page, the filters on the left side of the results can be combined to match the trusted repository criteria: under PID systems, select DOI (or hdl) to limit results to repositories that assign persistent identifiers; under Versioning, select yes; and under Data access, select open for the public synthetic dataset (or restricted when looking for a home for the original data). The Subjects filter narrows the list to a discipline (e.g., Social and Behavioural Sciences > Psychology), and a platform name can be typed into the search box to check a specific repository. Each repository's entry lists its identifier system, versioning policy, access conditions, and licensing, so these details can be confirmed before depositing (see Figure 10).

Discussion

The TOP 2025 revision clarifies a distinction that the original framework left unclear as transparency and verification are related but separate editorial functions (Center for Open Science, 2025; Grant et al., 2026). Disclosure requirements govern whether the materials underlying a claim are accessible, while verification requirements govern whether those materials are sufficient to independently evaluate the claim. Journals that formalize this distinction can set expectations for authors that are precise enough to implement consistently, rather than relying on broad commitments to openness that leave compliance underspecified (Naaman et al., 2023; Toomey et al., 2025). For *BRM*, aligning editorial policy with TOP 2025 extends practices that have shaped the journal's publication model since its founding (Brysbaert et al., 2021), giving those practices a shared vocabulary that

530 authors, reviewers, and other journals can recognize.

531 The synthetic data tutorial addresses a specific gap that common arises in submission
532 when disclosure is expected but the original data cannot be shared. Consent restrictions,
533 institutional review board limitations, and third-party data use agreements are common in
534 behavioral and social science research, and they do not disappear because a journal adopts
535 stronger transparency standards (Lee et al., 2018). Synthetic data generation gives authors a
536 way to meet the spirit of a data-sharing requirement. These resources allow others to inspect
537 the data structure, run the published analytic code, and confirm that the reported results
538 follow from the documented procedure without releasing identifiable participant records. The
539 **synthpop** (Nowok et al., 2016) and **jaspSyntheticData** tutorials demonstrated that this
540 approach can reproduce the pattern and magnitude of effects from the original analysis using
541 either a code-based or a point-and-click workflow, so the approach does not require
542 researchers to adopt new software environments to benefit from it.

543 Synthetic data are not a universal substitute for original data. As noted in the tutorial,
544 synthetic datasets are well suited to code verification, exploratory replication of a published
545 model, and methods development, but they are poorly suited to analyses where
546 individual-level heterogeneity or the exact original values are the scientific target, such as
547 meta-analyses or secondary analyses that depend on precise raw scores. Authors and
548 reviewers should treat synthetic data as one tool among several for reconciling data-sharing
549 expectations with confidentiality constraints (Grant et al., 2026), and should select the tool
550 that fits the sensitivity of the data and the intended use of the shared materials.

551 Several practical steps improve the long-term value of shared synthetic data. Running
552 a disclosure risk check before release, documenting the synthesis procedure and software
553 version, and depositing versioned updates as the synthesis or underlying data changes all
554 help ensure that a synthetic dataset remains interpretable and trustworthy over time (Nowok
555 et al., 2016). These steps are lightweight relative to the alternative of withholding data

556 entirely, and they extend naturally to other journals adopting TOP 2025 or comparable
557 transparency frameworks (Toomey et al., 2025).

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Table 1

TOP 2015 transparency standards

Area	Level 0
Data citation	No mention of data citation.
Data transparency	Data sharing is encouraged, or not ment
Analytical code transparency	Code sharing is encouraged, or not ment
Materials transparency	Materials sharing is encouraged, or not m
Reporting guidelines AKA Design and Analysis Transparency	No mention of reporting guidelines.
Study preregistration	Journal says nothing
Analysis plan preregistration	Journal says nothing
Replication	Journal says nothing
Publication Bias	Journal says nothing
Open science badges	

Table 2
Research Practices

Practice	Level 1: Disclosed
Study Registration	Authors stated whether or not a study was registered—and, if so, where a
Study Protocol	Authors stated whether or not the study protocol is available—and, if so,
Analysis Plan	Authors stated whether or not the analysis plan is available—and, if so, w
Materials Transparency	Authors stated whether or not materials are available—and, if so, where.
Data Transparency	Authors stated whether or not data are available—and, if so, where.
Analytic Code Transparency	Authors stated whether or not analytic code is available—and, if so, wher
Reporting Transparency	Authors stated whether or not they used a reporting guideline—and, if so

Table 3
Comparison of TOP Guidelines (2015 vs. 2025)

Feature	
Overall, Purpose	Focused on improving transparency in journal publications through structur
Organization of Standards	8 separate standards: Citation, Data Transparency, Analytic Code Transpa
Levels of Implementation	Level 0: Not addressed or Adopted; Level 1: Disclosure; Level 2: Requireme
Citation	Treated as its own standalone standard (citation of data, materials, and coo
Preregistration	Two standards: preregistration of studies and preregistration of analysis pla
Design & Analysis	Included as the Design and Analysis Transparency standard.
Verification	Implied verification appears only at Level 3 within each standard (e.g., inde
Replication	Replication was one of the eight standards; Registered Reports were treated
Intended Users	Designed primarily for journals.
Flexibility	Encouraged modular adoption. So, journals choose which standards and lev

Table 4

Comparison of regression coefficients from original and synthetic data

Term	B	SE	p	B	SE	p
(Intercept)	0.62	0.28	0.03	0.23	0.28	0.42
util_score	0.93	0.07	0.00	0.99	0.07	0.00
RegionSouthern	0.58	0.36	0.11	0.65	0.35	0.07
RegionWestern	0.91	0.36	0.01	0.88	0.35	0.01
Age	-0.02	0.00	0.00	-0.01	0.00	0.00
GenderFemale	-0.25	0.07	0.00	-0.16	0.07	0.03
util_score:RegionSouthern	-0.17	0.10	0.08	-0.21	0.09	0.03
util_score:RegionWestern	-0.19	0.10	0.05	-0.18	0.09	0.05

Note. Original (left) and synthetic (right) estimates. The model predicts moral acceptability of the trolley dilemma from utilitarian attitudes, world region, their interaction, age, and gender.

Table 5

*Comparison of regression coefficients from original data and
jaspSyntheticData synthetic data*

Term	B	SE	p	B	SE	p
(Intercept)	0.62	0.28	.029	0.57	0.29	.047
util_score	0.93	0.07	< .001	0.90	0.07	< .001
RegionSouthern	0.58	0.36	.111	0.58	0.37	.111
RegionWestern	0.91	0.36	.011	0.86	0.36	.018
Age	-0.02	0.00	< .001	-0.01	0.00	< .001
GenderFemale	-0.25	0.07	< .001	-0.22	0.07	.003
util_score:RegionSouthern	-0.17	0.10	.076	-0.15	0.10	.122
util_score:RegionWestern	-0.19	0.10	.048	-0.13	0.10	.179

Note. Original (left) and JASP synthetic (right) estimates.

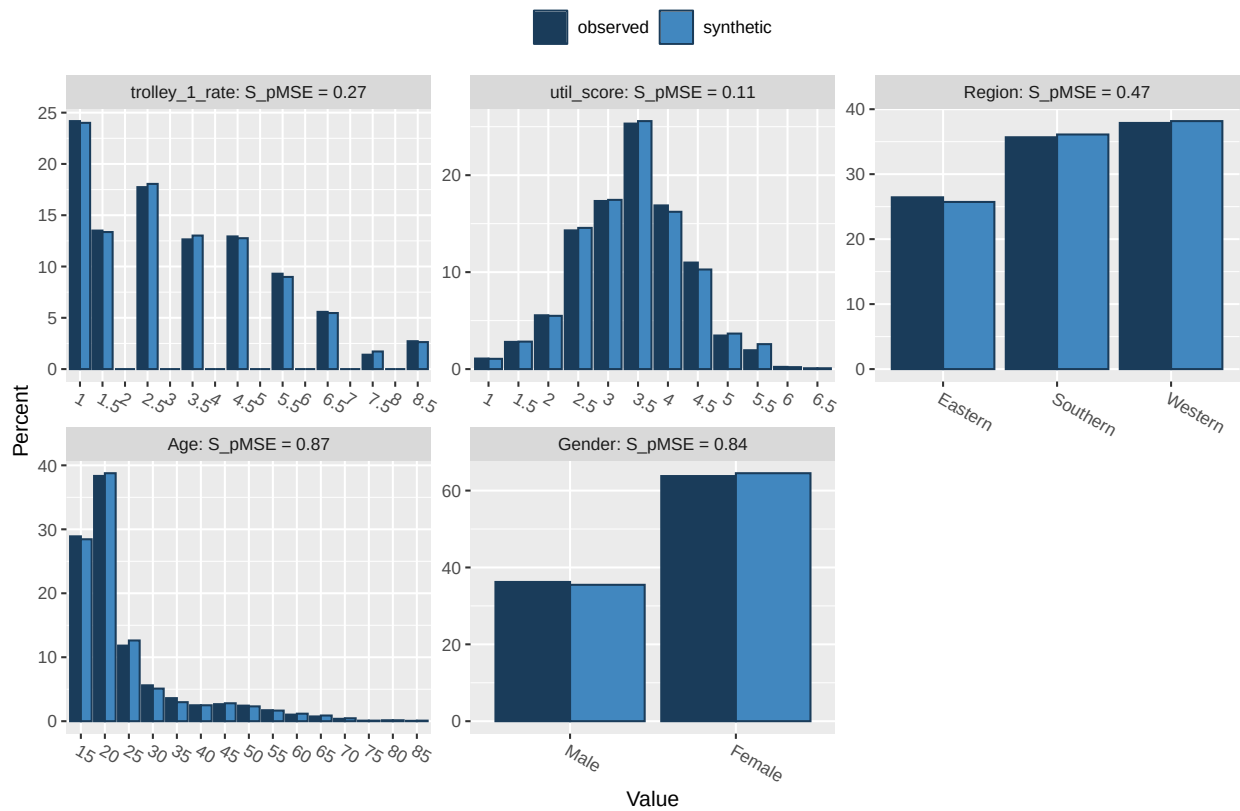


Figure 1. Distribution comparison between original and synthetic data for each analytic variable.

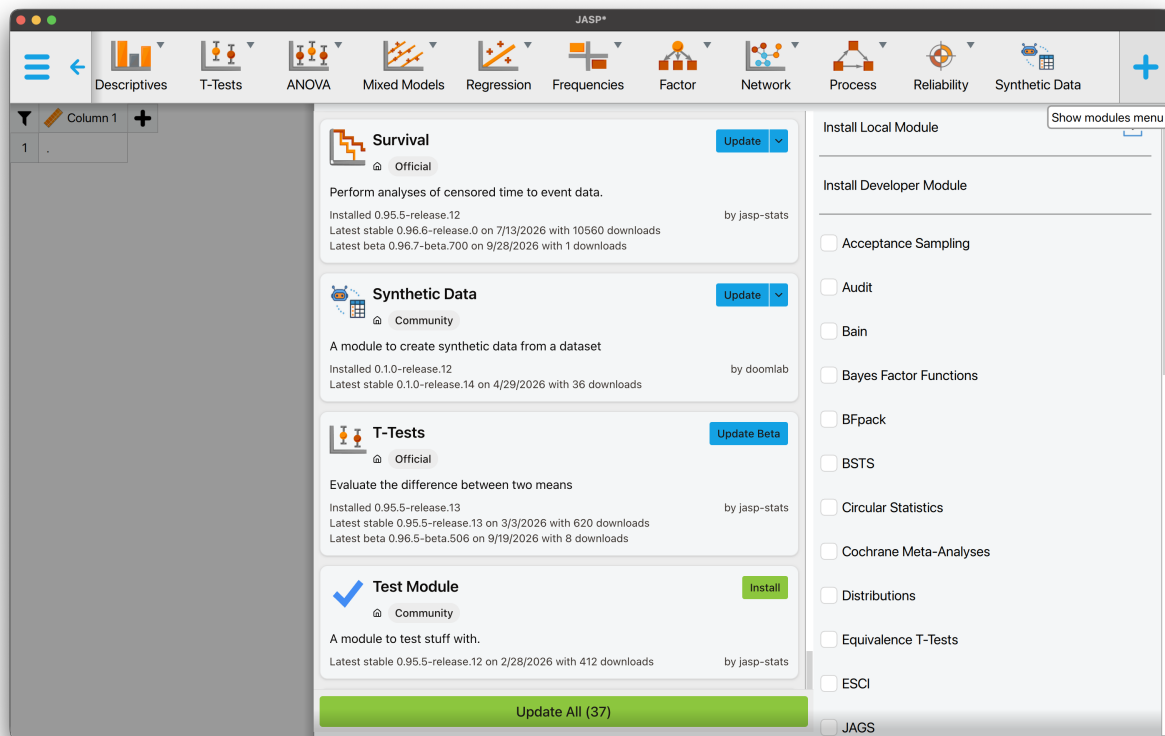


Figure 2. JASP module menu showing the jaspSyntheticData module available for installation. The Modules button (top right) opens the module library where jaspSyntheticData can be found and added to the interface.

	survey_name	SV_OGO7CYZ1zsBNK2F	Date	EndDate	Responseld	trolley_1_resp	trolley_1_r
1	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-15T23:06:13Z	2020-04-15T23:17:26Z	R_1jDEG6ytkgDCcuf	.	.
2	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:11:49Z	2020-04-17T03:29:08Z	R_2RU2UpJ8zh5Yy5	.	.
3	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:18:34Z	2020-04-17T03:29:49Z	R_2sRysW6J8VxFirA	.	.
4	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:21:06Z	2020-04-17T03:31:41Z	R_pRVoGkzZGLg8WQh	1	4
5	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:23:35Z	2020-04-17T03:35:01Z	R_eaONvj9VEFSB43f	.	.
6	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:15:58Z	2020-04-17T03:35:22Z	R_wOgUikKg9aJ6NFv	2	3
7	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:38:23Z	2020-04-17T03:47:02Z	R_3dLu9VWaHKmwT	2	3
8	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:24:48Z	2020-04-17T04:05:12Z	R_2YElH8Y4fx3a6Rb	2	2
9	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T04:00:32Z	2020-04-17T04:09:17Z	R_XhyHm6rUShA3L8Z	.	.
10	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:50:31Z	2020-04-17T04:10:05Z	R_3oBfKmf92mhcNSJ	2	3
11	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:50:57Z	2020-04-17T04:10:31Z	R_R49lSfLFoYOnNTP	.	.
12	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T03:31:01Z	2020-04-17T04:19:03Z	R_1HeQrRCdOJvKkyR	.	.
13	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T04:24:59Z	2020-04-17T04:38:06Z	R_3hyUb5SCz1feUQY	2	3
14	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T04:12:37Z	2020-04-17T04:39:59Z	R_3G3UO5idhDCEFXU	2	2
15	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T04:48:31Z	2020-04-17T04:57:41Z	R_9XZ0y84BE7ZXOBr	.	.
16	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T04:11:58Z	2020-04-17T05:01:25Z	R_bjgwI9J7SPUGjCx	1	9
17	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T04:49:04Z	2020-04-17T05:02:47Z	R_1PRjdISYLnrtbBP	1	8
18	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T05:25:12Z	2020-04-17T05:42:57Z	R_Zlf4wLQ3hYBCJ5D	2	3
19	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T06:08:19Z	2020-04-17T06:21:05Z	R_3Ld64Vc9jCzthPL	2	3
20	PSA006_Eastern	SV_OGO7CYZ1zsBNK2F	2020-04-17T06:11:27Z	2020-04-17T06:31:47Z	R_302J848oZXwkhk	.	.

Figure 3. The trolley analytic dataset loaded in JASP's spreadsheet view. Continuous variables (trolley_1_rate, util_score, Age) are marked as scale; Region and Gender are marked as nominal. Variable types can be adjusted using the icons in the column headers.

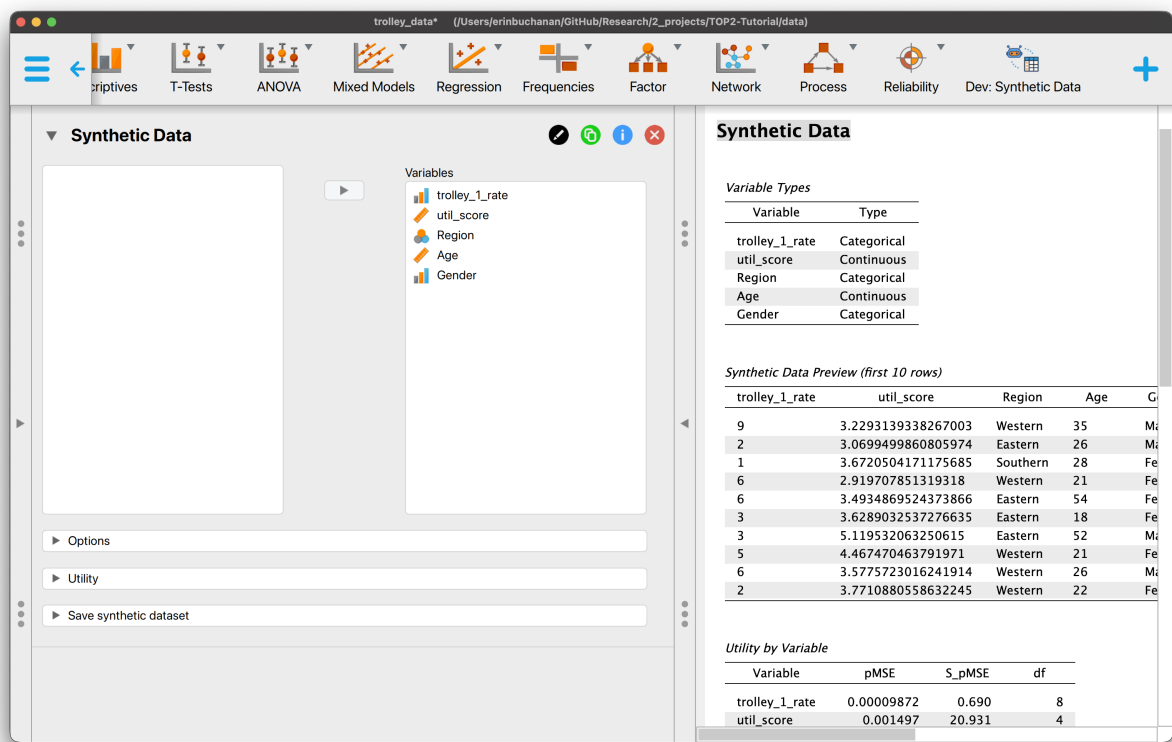


Figure 4. The jaspSyntheticData analysis panel with all five analytic variables assigned. The synthesis order follows the variable list from top to bottom.

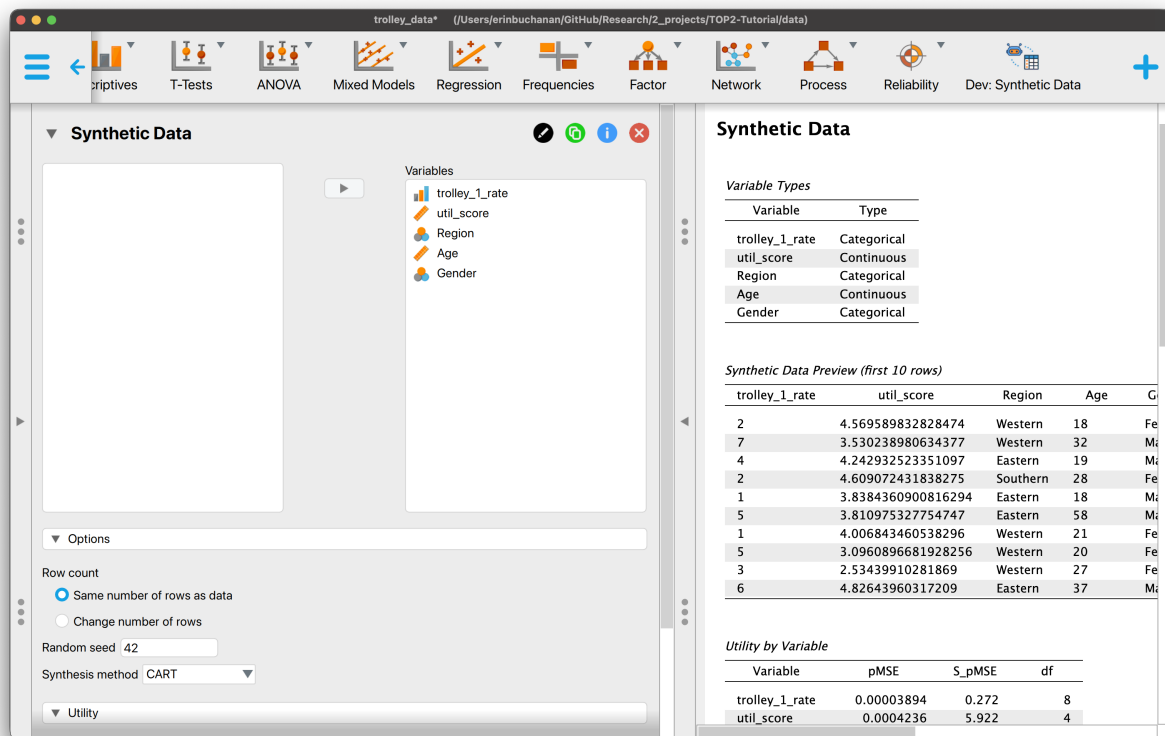


Figure 5. The Options section in jaspSyntheticData showing the row count, random seed, and synthesis method (CART) settings. A seed value of 42 is entered to match the *R* tutorial.

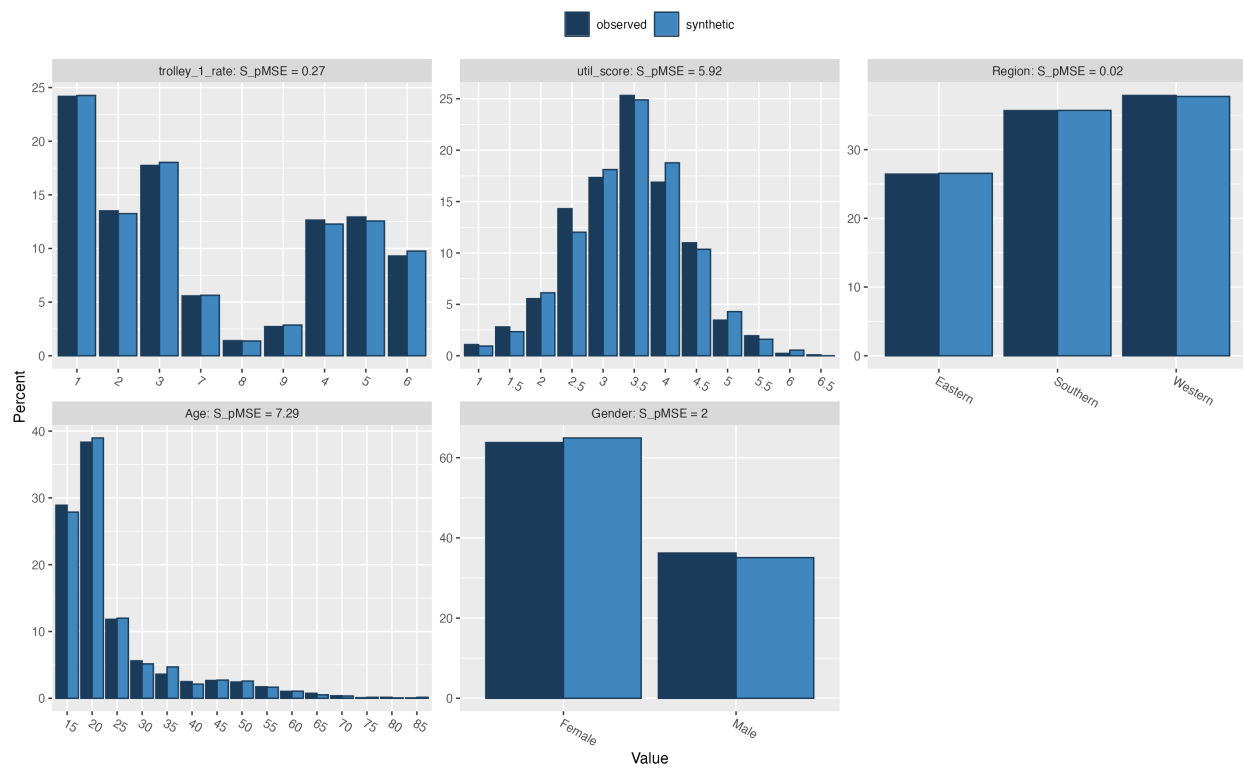


Figure 6. Distribution comparison plots from jaspSyntheticData for all five analytic variables. Original data are shown in dark blue; synthetic data are shown in light blue. Each panel title reports the standardized pMSE for that variable.

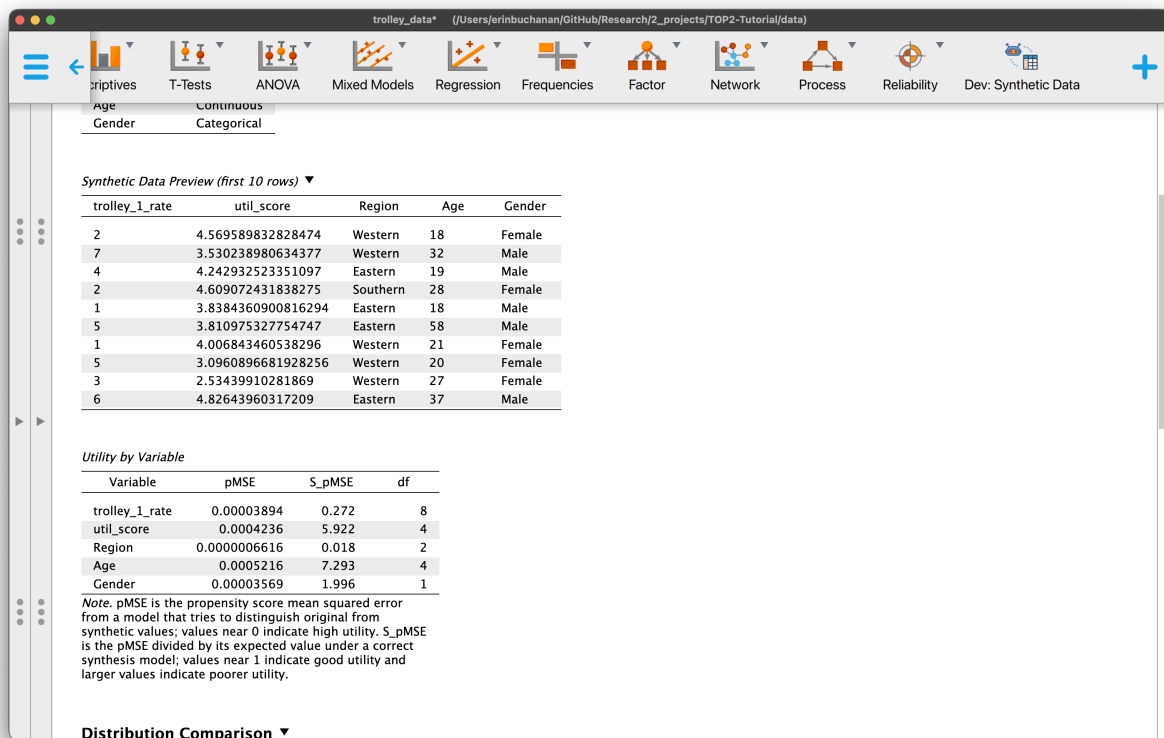


Figure 7. Utility by Variable table from jaspSyntheticData, showing the pMSE and standardized pMSE for each variable.

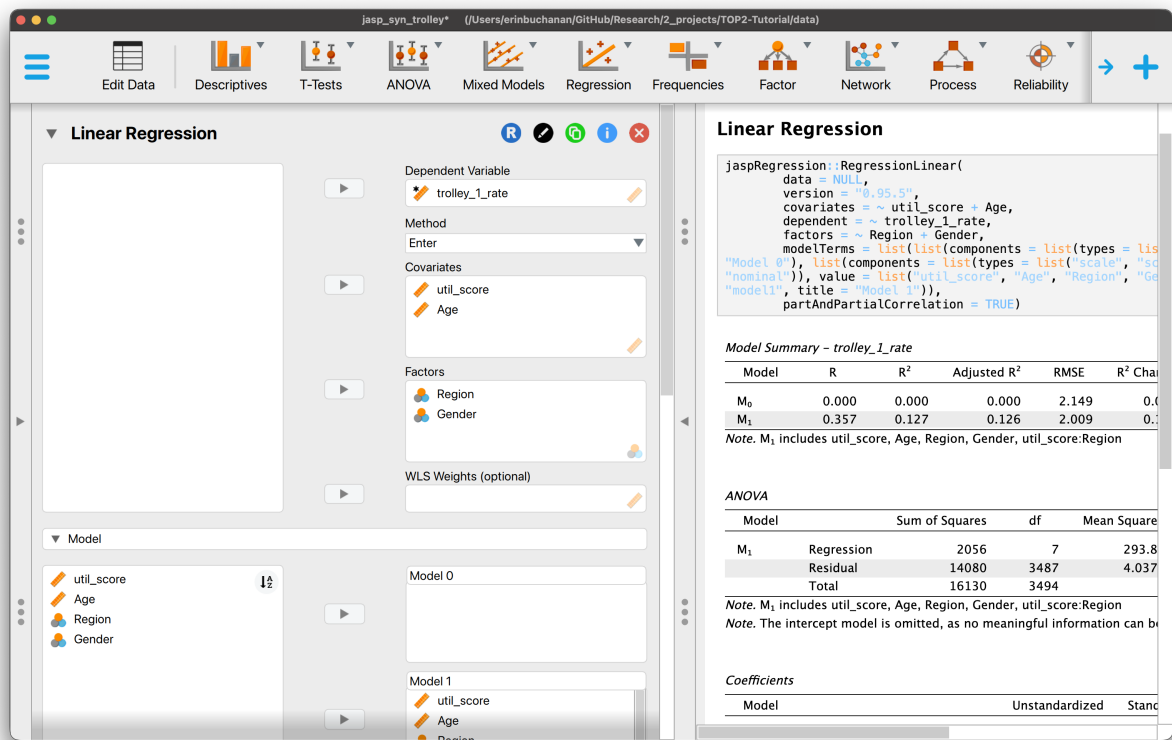


Figure 8. Linear regression analysis conducted on the saved synthetic dataset in JASP. The coefficient estimates are compared with those from the original analysis in Table reftab:jasp-model-comparison.

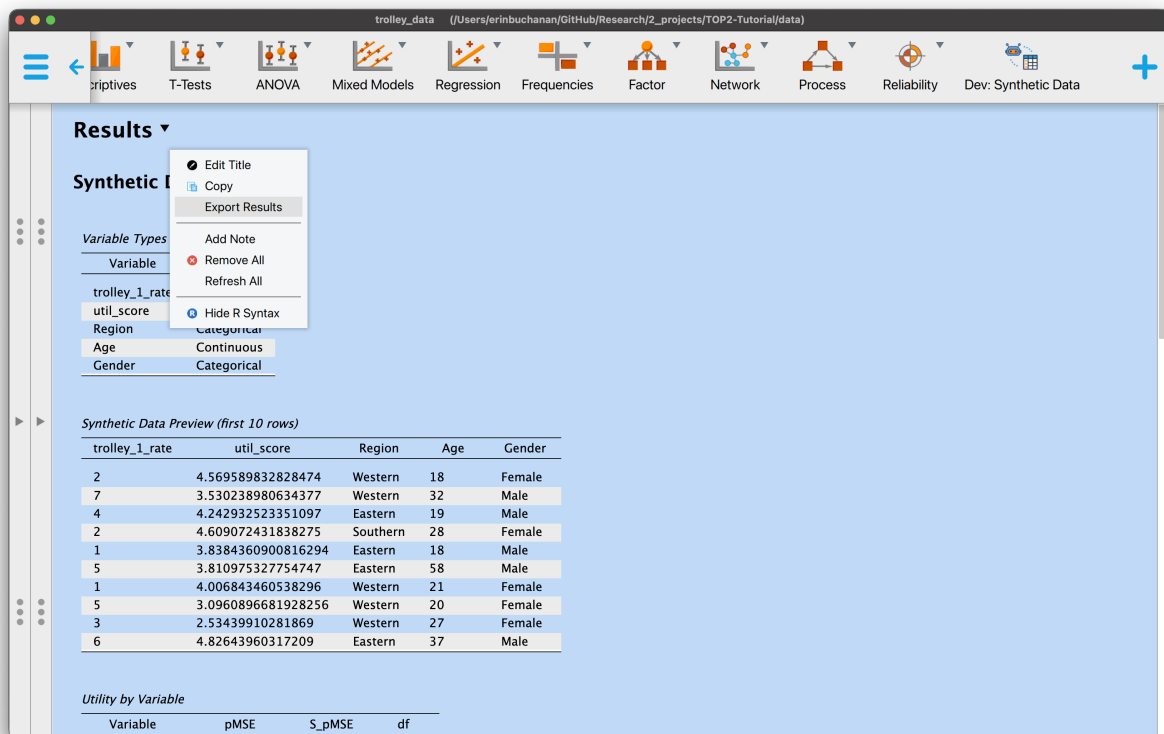


Figure 9. Exporting JASP results as an .html file (File > Export Results). Unlike the .jasp analysis file, the exported results include the output and analysis settings but not the original data.

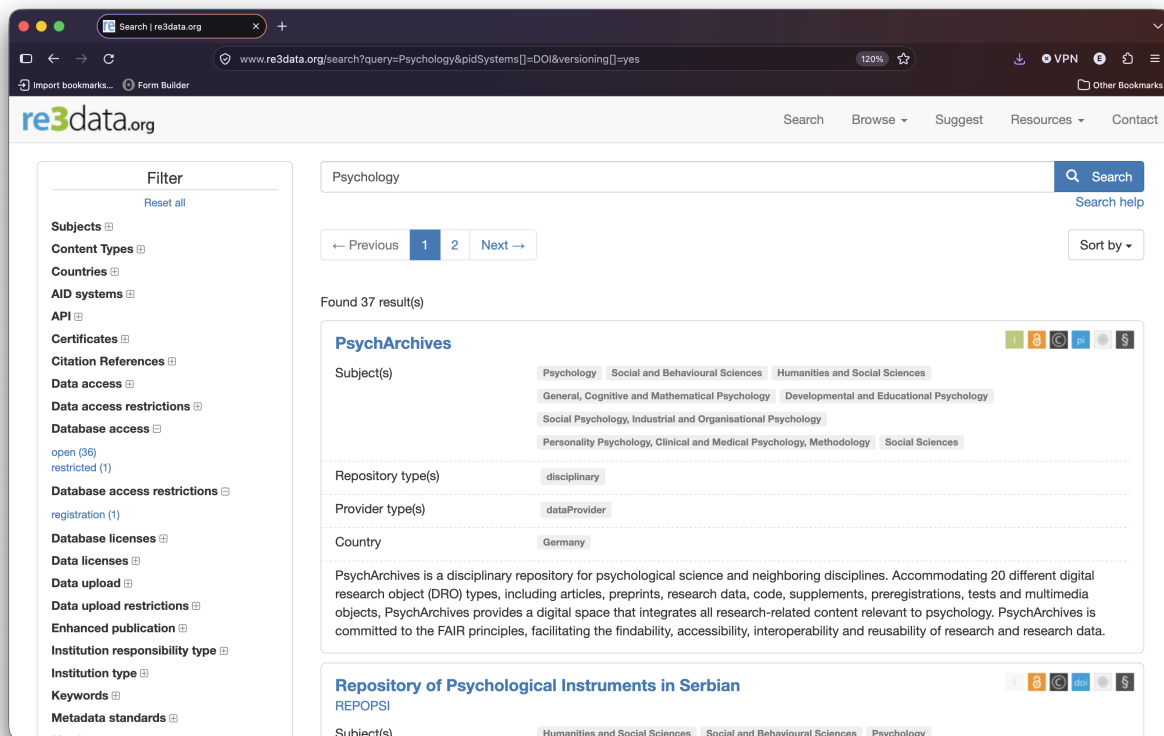


Figure 10. Searching re3data.org for trusted repositories. Filters on the left limit results to repositories that assign persistent identifiers (PID systems: DOI), support versioning (Versioning: yes), provide open data access, and cover the relevant subject area.